

X'. Give the coordinates as 215 and 175, with a duration of 5 minutes. Set the repeat count at 5, as shown in Figure 6. Select the option to restore the original value, so that at the end of the animation the car comes to its original position. A time delay can be created in between the movements of the car. The operation in the SVG graphics node is now complete.

Connect the button node from the exit side of the SVG graphics node. On double-clicking, the *Properties* dialog box will appear. Set the group as 'project1' and tab as 'automation', as shown in Figure 7. The colour of the button can be modified. Any message can be given to the *msg.payload* option, as shown in Figure 7.

After setting up the button node, connect the inject node to the SVG graphics node. The desired input for the movement of the car can be given, so that the car moves according to the given instructions. Double-click the inject node to change its properties. The payload messages that can be given are 'start the car', 'move backward' and 'move forward'. The animation can be made to repeat after a certain interval (5 minutes, as shown in Figure 8). Another choice is to trigger the start of the flow at a certain time every day. The audio out node is added to the flow and is wired to the button node. The group and tab are automatically set because of the connection we made between the nodes. An option is provided to choose the desired language and gender of the voice for the output, as shown in Figure 8.

After finishing the connections depicted in Figure 2, the Node-RED dashboard (output window) can be viewed by opening the URL <http://127.0.0.1:1880/ui> in a browser.

In the output window, the car is initially in a stationary position, as shown in Figure 9. Once the button is pressed, the car will move back and

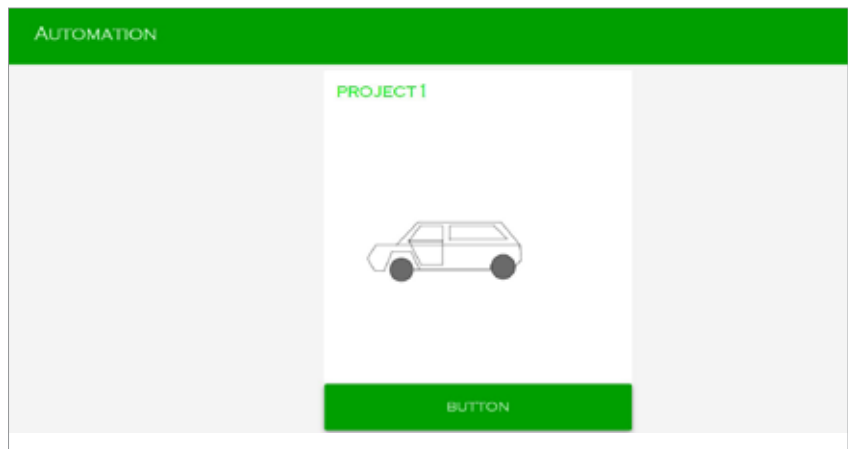


Figure 9: Node-RED dashboard showing the output

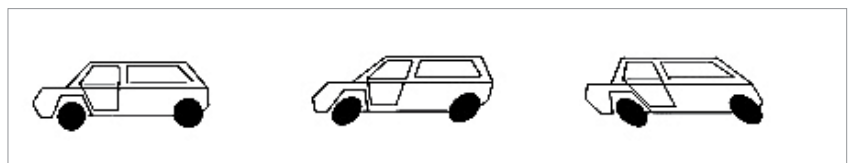


Figure 10.1: Stationary car

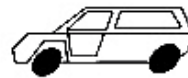


Figure 10.2: Forward movement of the car



Figure 10.3: Backward movement of the car

forth according to the values specified. This can be seen in Figure 10.

Autonomous cars can be of great help in critical locations like nuclear power plants, atomic centres, huge data centres, industrial applications with

extremely high or low temperatures, and space centres. They are a great boon for the elderly and visually impaired. But they need to be designed with the utmost care to ensure there are no accidents. **END** 🐧

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